BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023 • Paper Code: 6475

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Physics - Paper: I - Objective (iii)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6475

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Note: SECTION-A

Q1. The refractive index of water is 1.33 the speed of light in water is: ($c = 3x10^8 m/sec$)

(A) $3x10^8 m/sec$
(B) $1.8x10^8 m/sec$
(C) $2.3x10^8 m/sec$
(D) $1.3x10^8 m/sec$

Q2. In Michelson interferometer, to switch the fringes from bright to dark, the mirror should be displaced by:

(A) $\lambda/2$
(B) $\lambda/3$
(C) $\lambda/4$
(D) λ

Q3. Speed of sound at 0°C is 332 ms^{-1}. The speed of sound at 30°C will be:

(A) 332 m/sec
(B) 350 m/sec
(C) 340 m/sec
(D) 335 m/sec

Q4. A block weighing 4.0 kg extends a spring by 0.16m from its unstretched position, stretching force is:

(A) 20.3 N
(B) 16.3 N
(C) 39.2 N
(D) 14.2 N

Q5. A simple harmonic oscillator has a period of 0.01s and an amplitude of 0.2m. The magnitude of velocity at the centre of oscillation is:

(A) 20π
(B) 40π
(C) 60π
(D) 80π

Q6. The dimensions of $\frac{1}{2}\rho v^2$ is same as that of:

(A) K.E
(B) P.E
(C) Pressure
(D) Work

Q7. Which electromagnetic wave is used for the satellite communication system?

(A) Radio waves
(B) Infrared waves
(C) Ultraviolet waves
(D) Microwaves

Q8. 2° is equal to:

(A) 0.035 rad.
(B) 0.300 rad.
(C) 0.350 rad.
(D) 0.0035 rad.

Q9. An electric motor produces a tension of 4500N in a load lifting cable and rolls it at the rate of $2ms^{-1}$. The power of the motor is:

(A) 4kW
(B) 2kW
(C) 15kW
(D) 9kW

Q10. Acceleration of rocket is given by relation:

(A) a = M/mv
(B) a = m/Mv
(C) a = mv/m
(D) a = mv/M

Q11. Speed of the earth around the sun in ms^{-1} is:

(A) 35500
(B) 20000
(C) 29600
(D) 50000

Q12. Two forces of magnitudes 10N and 20N act on a body in directions making angles of 30° each with x-axis. The x-component of the resultant force will be:

(A) 25.98 N
(B) 30.98 N
(C) 20.98 N
(D) 17.98 N

Q13. If $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$ then angle between $\vec{a}$ and $\vec{b}$ is:

(A) 0°
(B) 45°
(C) 90°
(D) 180°

Q14. Which pair of physical quantities have same dimensions?

(A) Work and power
(B) Momentum and impulse
(C) Force and torque
(D) Momentum and force

Q15. Velocity of an object has 1% uncertainty and mass has 2% uncertainty then total uncertainty in K.E will be:

(A) 3%
(B) 2%
(C) 4%
(D) 1%

Q16. The change in internal energy is defined as:

(A) Q − T
(B) Q + P
(C) Q − P
(D) Q − W

Q17. Absolute zero corresponds to:

(A) −459°F
(B) −360°F
(C) 0°F
(D) 460°F

Physics - Paper: I - Subjective

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note: Section B is compulsory. Attempt any Three questions from Section C.

Note: SECTION-B

Q2.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Distinguish between base units and derived units.

ii. What are random error and systematic error?

iii. Write the dimensions of: (i) Pressure (ii) Power.

iv. The period of a simple pendulum is measured by a stop watch. What type of errors are possible in the time period?

v. Define: (i) Null vector (ii) Equal vectors.

vi. Vector $\vec{A}$ lies in the xy-plane. For what orientation will both of its rectangular components be negative? For what orientation will its components have opposite signs?

vii. Can a vector have component greater than the vector's magnitude?

viii. Water flows out from a pipe at $3kgs^{-1}$ and its velocity changes from $5ms^{-1}$ to zero on striking the wall. What will be the force exerted by water on the wall?

ix. Derive formula for time of flight of projectile.

x. What is the difference between elastic and inelastic collision?

xi. What is the difference between uniform and variable velocity?

xii. Two row boats moving parallel in the same direction are pulled towards each other. Explain.

Q3.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. A girl drops a cup from a certain height which breaks into pieces. Which energy changes are involved?

ii. In which case more work is done? When a 50 kg bag of books is lifted through 50cm or when a 50 kg crate is pushed through 2m across a force of 50N?

iii. Potential energy is the property of a conservative field. Explain.

iv. What is geostationary satellite and geo-stationary radius?

v. Show that orbital angular momentum $L_o = mvr$

vi. What is meant by moment of inertia? Explain its significance.

vii. If a mass spring system is hung vertically and set into oscillation, why does the motion eventually stop?

viii. Show that when a pendulum moves from mean position to half of amplitude, time taken by it is $t = T/12$.

ix. What is meant by phase angle?

x. Explain 'red shift' and 'blue shift' in light.

xi. Why does sound travel faster in solids than in gases?

xii. How are beats useful in tuning musical instruments?

Q4.. (Marks 12) Write short answers to any Six parts. (6 x 2 = 12)

i. State Huygen's principle.

ii. How would you manage more orders of spectra by using a diffraction grating?

iii. How is the distance between interference fringes affected by the separation between the slits of Young's experiment? Can fringes disappear?

iv. What is the least distance of distinct vision? Also, give the length of this distance.

v. If a person was looking through a telescope at the full moon, how would the appearance of the moon be changed by covering half of the objective lens?

vi. What is the average translational K.E of molecules in a gas at temperature 27°C?

vii. How would you explain the sign convention of First Law of Thermodynamics?

viii. Why is the average velocity of the molecules in a gas zero but the average of the square of velocities is not zero.

ix. Why does the pressure of a gas in a car tyre increase when it is driven through some distance?

> Note: SECTION-C (Note: Attempt any Three questions. Each question carries Eight (8) Marks)

Q5.. (Marks 8) (a) What is gravitational field? Show that gravitational field is a conservative field. (5)

(b) Find the projection of vector $\vec{A} = 2\hat{i} - 8\hat{j} + \hat{k}$ in the direction of the vector $\vec{B} = 3\hat{i} - 4\hat{j} - 12\hat{k}$. (3)

Q6.. (Marks 8) (a) State and prove the law of conservation of linear momentum. (5)

(b) A body of moment of inertia I = 0.80 kg m^2 about a fixed axis, rotate with a constant angular velocity of 100 rad s^{-1}. Calculate its angular momentum and the torque to sustain this motion. (3)

Q7.. (Marks 8) (a) State and prove Bernoulli's equation for an ideal fluid. (5)

(b) A carnot engine whose low temperature reservoir is at 7°C has an efficiency of 50%. It is desired to increase the efficiency to 70%. By how many degrees the temperature of the source be increased? (3)

Q8.. (Marks 8) (a) Define simple harmonic motion. Discuss that energy is conserved for a body executing simple harmonic motion. (5)

(b) Find the temperature at which the velocity of sound in air is two times its velocity at 10°C. (3)

Q9.. (Marks 8) (a) Describe how Michelson measured the speed of light? (5)

(b) In a double slit experiment the second order maximum occurs at θ = 0.25°. The wavelength is 600 nm. Determine the slit separation. (3)

GROUP – II

Physics - Paper: I - Objective (iii)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6476

> Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

> Note: SECTION-A

Q1. Angular momentum has the same unit as:

(A) Impulse x Distance
(B) Power x Time
(C) Linear momentum
(D) Work x Frequency

Q2. If a gymnast sitting on a stool with his arm stretched, lowers his arms:

(A) ω decreases
(B) ω increases
(C) I increases
(D) ω remains same

Q3. The working of carburetor of car uses:

(A) Equation of continuity
(B) Gravitational law
(C) Stoke's law
(D) Bernoulli's theorem

Q4. Which expression is correct for time period of simple pendulum?

(A) $T \propto l$
(B) $T \propto \sqrt{l}$
(C) $T \propto m$
(D) $T \propto g$

Q5. With the increase of temperature, speed of sound:

(A) Remains same
(B) Becomes zero
(C) Decreases
(D) Increases

Q6. Beats can be heard when difference of frequency is not more than:

(A) 4 Hz
(B) 6 Hz
(C) 8 Hz
(D) 10 Hz

Q7. Newton's rings are formed due to:

(A) Diffraction
(B) Refraction
(C) Reflection
(D) Interference

Q8. Information carrying capacity of an optical fibre is called:

(A) Capacity
(B) Band width
(C) Immunity
(D) Ability

Q9. No entropy change takes place in:

(A) Isobaric process
(B) Isothermal process
(C) Adiabatic process
(D) Isochoric process

Q10. If heat is added to a system, then its entropy will:

(A) Increase and positive
(B) Decrease and positive
(C) Increase but negative
(D) Decrease but negative

Q11. The earth receives larger amount of energy from:

(A) Wind (B) Water

(C) Sun (D) Moon

Q12. At what speed, the momentum and kinetic energy of a body have the same value:

(A) $1ms^{-1}$ (B) $2ms^{-1}$

(C) $4ms^{-1}$ (D) $8ms^{-1}$

Q13. Which of the given variable is present in all equations of motion:

(A) Distance (B) Acceleration

(C) Time (D) Torque

Q14. Product $\hat{i} \times (\hat{j} \times \hat{k})$ is equal to:

(A) 0 (B) $\vec{0}$

(C) $\hat{j}$ (D) $\hat{i}$

Q15. Projection of $\vec{A}$ on $\vec{B}$ is:

(A) A cos θ (B) B sin θ

(C) B tan θ (D) A sin θ

Q16. Absolute uncertainty in a measuring instrument is equal to:

(A) Least count (B) Accuracy

(C) Fractional uncertainty (D) Percentage uncertainty

Q17. The dimensional formula for the quantity light year is:

(A) [T] (B) [L]

(C) $[LT^{-1}]$ (D) [M°LT]

Physics - Paper: I - Subjective

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note: Section B is compulsory. Attempt any 3 questions from Section C.

Note: SECTION-B

Q2.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Show that $E+mc^2$ is dimensionally consistent.

ii. The time of 30 vibrations of a simple pendulum recorded by a stop watch accurate up to one tenth of a second is 54.6 s. Find its period with uncertainty.

iii. Give the drawbacks to use the period of a pendulum as a time standard.

iv. What is the difference between kilogram and mole?

v. Can the product of two vectors be equal to the product of their magnitudes?

vi. Define the terms (i) unit vector and (ii) components of a vector.

vii. If $\vec{A} + \vec{B} = \vec{0}$, what can you say about the components of the two vectors?

viii. Define impulse and write its units.

ix. Show that the range of projectile is maximum when projectile is thrown at an angle of 45° with the horizontal.

x. Can the velocity of an object reverse the direction when acceleration is constant? If so give an example.

xi. Derive the relation for the time of flight of a projectile.

xii. Explain, how the swing is produced in a fast moving cricket ball.

Q3.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. A boy uses a catapult to throw a stone which accidentally smashes a greenhouse window. List the possible energy changes.

ii. An object has a 1 J of potential energy. Explain what does it mean?

iii. What do you know about geothermal energy?

iv. Why does a diver change his body positions before and after diving in the pool?

v. Show that orbital angular momentum $L_0 = mvr$.

vi. What is INTELSAT VI?

vii. If a mass spring system is hung vertically and set into oscillations, why does the motion eventually stop?

viii. What is meant by phase angle? Does it define angle between maximum displacement and the driving force?

ix. Define restoring force. What are its units?

x. How should a sound source move with respect to an observer so that the frequency of its sound does not change?

xi. Why does sound travel faster in solids than in gases?

xii. How are beats useful in tuning musical instruments?

Q4.. (Marks 12) Write short answers to any Six parts. (6 x 2 = 12)

i. In Young's experiment, one of the slits is covered with blue filter and other with red filter. What would be the pattern of light intensity on the screen?

ii. How would you distinguish between un-polarized and plane-polarized lights?

iii. Give two uses of Michelson's interferometer.

iv. If a person was looking through a telescope at the full moon, how would the appearance of the moon be changed by covering half of the objective lens?

v. What is spectrometer? Write down its two main parts.

vi. Given an example of a natural process that involves an increase in entropy.

vii. What happens to the temperature of a room, when an air conditioner is left running on a table in the middle of the room?

viii. Define Iso-thermal process and write its relation.

ix. In which process external work is done at the expense of the internal energy of the gas molecules? Explain it.

> Note: SECTION-C (Note: Attempt any Three questions. Each question carries Eight (8) Marks)

Q5.. (Marks 8) (a) Define gravitational field. Show that work done in gravitational field is independent of path followed. (5)

(b) Two forces of magnitude 10 N and 20 N act on a body in directions making angles 30° and 60° respectively with x-axis. Find the resultant force. (3)

Q6.. (Marks 8) (a) What is projectile and projectile motion? Explain and analyse the oblique projectile as well as horizontal projectile. (5)

(b) What is the least speed at which an aeroplane can execute a vertical loop of 1.0 km radius so that there will be no tendency for the pilot to fall down at the highest point. (3)

Q7.. (Marks 8) (a) Show that pressure exerted by the gas molecules is directly proportional to average translational kinetic energy of the gas molecules. (5)

(b) Water flows through a hose, whose internal diameter is 1cm at a speed of 1 ms^{-1}. What should be the diameter of the nozzle if water is to emerge at 21 ms^{-1}? (3)

Q8.. (Marks 8) (a) What is the main purpose of resonance? Explain it with an experiment of pendulums of different lengths. Also, give one example of resonance. (5)

(b) A stationary wave is established in a string which is 120 cm long and fixed at both ends. The string vibrates in four segments at a frequency of 120 Hz. Determine its wavelength and the fundamental frequency? (3)

Q9.. (Marks 8) (a) What is a compound microscope? Describe its construction and working. Also calculate its magnifying power. (5)

(b) A light is incident normally on a grating which has 2500 lines per centimetre. Compute the wavelength of a spectral line for which the deviation in second order is 15.0°. (3)